PROJECT REPORT

TASK 29: RELATIONSHIP ANALYSIS USING PYTHON

1. Abstract

This project focuses on analyzing the relationship between two numerical variables using Python, Pandas, NumPy, and Matplotlib.

A student performance dataset containing study hours and examination marks is used to create a scatter plot and identify the observed relationship.

Pearson correlation is calculated to understand the direction and strength of the linear association between the variables.

The project demonstrates how data visualization and correlation analysis can help identify patterns in numerical data.

2. Introduction

Relationship analysis is an important part of exploratory data analysis.

It helps understand how two numerical variables are associated with one another.

A scatter plot is a graphical representation that displays individual observations as points on a two-dimensional coordinate system.

The pattern of these points helps identify positive, negative, or weak relationships.

This project analyzes the relationship between the number of hours students spend studying and the marks they obtain in examinations.

3. Objectives

The main objectives are:

- To create a scatter plot using two numerical variables.
- To understand the relationship between study hours and examination marks.
- To calculate Pearson correlation.
- To identify the direction and strength of the relationship.
- To interpret the observed pattern.
- To record at least three observations from the visualization.

4. Tools and Technologies

Python: Used to implement the relationship analysis.

Pandas: Used to create, load, and process the dataset.

NumPy: Used for numerical operations and calculations.

Matplotlib: Used to create the scatter plot and trend line.

Google Colab: Used as the Jupyter Notebook development environment.

CSV: Used to store the student dataset and analysis results.

5. Dataset Description

The project uses a manually created student performance dataset containing 20 fictional student records.

The dataset includes three columns:

Column	Description
student_id	Unique student identifier
study_hours	Hours spent studying
exam_marks	Marks obtained in the examination

The dataset is designed to demonstrate a positive relationship between study time and examination performance.

It is created for educational purposes and does not represent actual examination results.

6. Methodology

The project follows these steps:

1. Import Pandas, NumPy, and Matplotlib.
2. Create a sample student performance dataset.
3. Convert the data into a Pandas DataFrame.
4. Export and reload the CSV dataset.
5. Inspect dataset structure and data types.
6. Calculate descriptive statistical measures.
7. Create a scatter plot using Matplotlib.
8. Add a linear trend line to the visualization.
9. Calculate the Pearson correlation coefficient.
10. Identify the direction and strength of the relationship.

11. Generate observations from the results.
12. Export the dataset and relationship summary.

7. Scatter Plot Visualization

A scatter plot is created using Matplotlib's `plt.scatter()` function.

The visualization represents:

- X-axis: Study hours.
- Y-axis: Examination marks.
- Each point: One student observation.

The scatter plot helps identify the overall pattern between the two variables.

Insert the actual study_hours_vs_exam_marks.png chart here.

8. Correlation Analysis

Pearson correlation measures the strength and direction of a linear relationship between two numerical variables.

The correlation coefficient ranges from -1 to +1.

- +1 represents a perfect positive linear relationship.
- -1 represents a perfect negative linear relationship.
- 0 indicates no linear correlation.

The calculated Pearson correlation coefficient for this dataset is approximately:

Correlation = 0.9754

This indicates a strong positive linear association between study hours and examination marks.

9. Trend Line Analysis

A linear trend line is added to the scatter plot using NumPy's `polyfit()` function.

The upward direction of the trend line indicates that higher study hours are generally associated with higher examination marks in the sample dataset.

The trend line helps visualize the overall direction of the relationship.

Insert the actual scatter plot with the regression trend line here.

10. Statistical Summary

The analysis produces the following results:

Measurement	Value
Total students	20
Average study hours	5.75
Average examination marks	63.50
Minimum study hours	1
Maximum study hours	10.5
Minimum examination marks	35
Maximum examination marks	96
Pearson correlation	0.9754

These results help describe the numerical characteristics of both variables.

11. Key Observations

The scatter plot and correlation analysis provide the following observations:

1. The dataset shows a strong positive relationship between study hours and examination marks.
2. Students who studied for more hours generally obtained higher examination marks.
3. The Pearson correlation coefficient is approximately 0.9754, indicating a strong positive linear association.
4. Study hours range from 1 to 10.5 hours.
5. Examination marks range from 35 to 96.
6. The scatter plot shows an overall upward pattern, although individual students show small variations.

12. Importance of Relationship Analysis

Relationship analysis helps understand how two variables are associated.

Scatter plots provide a visual method for identifying patterns, while correlation coefficients provide a numerical measure of linear association.

These techniques are useful in exploratory data analysis and help analysts understand datasets before applying more advanced statistical or machine-learning methods.

13. Applications

Relationship analysis is useful in:

- Student performance analysis.
- Study time and examination performance analysis.
- Sales and advertising analysis.
- Salary and experience analysis.
- Housing price analysis.
- Business performance analysis.
- Exploratory data analysis.
- Machine-learning feature analysis.

14. Limitations

The project uses a small, manually created dataset of 20 student records.

The dataset is designed to demonstrate a positive relationship and does not represent actual student examination results.

Correlation does not establish causation. Other factors, such as teaching quality, previous knowledge, and learning methods, may also influence examination performance.

15. Future Enhancements

The project could be extended by:

- Using larger real-world student datasets.
- Analyzing multiple numerical variables.
- Comparing positive and negative relationships.
- Creating correlation heatmaps.
- Performing regression analysis.
- Studying relationships between additional academic factors.
- Developing predictive models for student performance.

16. Conclusion

This project demonstrates how Python, Pandas, NumPy, and Matplotlib can be used to analyze relationships between numerical variables.

A scatter plot is created to visualize study hours against examination marks, and Pearson correlation is calculated to measure the strength and direction of the relationship.

The sample dataset produces a correlation coefficient of approximately 0.9754, indicating a strong positive linear association.

The project provides practical experience in data visualization, correlation analysis, and interpreting relationships between numerical variables.

17. References

1. Pandas Documentation: https://pandas.pydata.org/docs/
2. NumPy Documentation: https://numpy.org/doc/
3. Matplotlib Documentation: https://matplotlib.org/stable/
4. Matplotlib Scatter Plot Reference: https://matplotlib.org/stable/api/_as_gen/matplotlib.pyplot.scatter.html

19. Project Links

GitHub Repository: https://github.com/venky1318/Relationship_Analysis